Project 3 Active Learning for Learning to Defer

Report

Human Centric Artificial Intelligence
9 September 2026

Project Overview

In this project, we developed a system capable of classifying a news article from AG News dataset on its own or deferring the decision to a simulated human expert. Task 1 is just the plain AI baseline. Task 2 is where we build an expert who is good at one thing(sports). Task 3, on the other hand, attempts to learn when it is better to trust the expert rather than the AI. Task 4 does the same thing, but with one difference: we can only ask the expert a limited number of questions throughout the process.

Our baseline (TF-IDF and Logistic Regression) achieved a test accuracy of 91.80%. Task 3 remained at 91.43%, and the best result we achieved from Task 4 was 91.39%. Thus, we made the entire workflow operational from beginning to end, but no decision-transfer system on its own was able to outperform simple AI.

Data and Evaluation

AG News has four categories: World, Sports, Business, and Sci/Tech. We trained on the full 120,000 training articles and only touched the 7,600 test articles for evaluation purposes. We report test accuracy for all four tasks. For Tasks 3 and 4, we also report how often the system deferred a decision and whether those deferrals beneficial or harmful compared to the AI alone, and what proportion of the deferred cases resulted in correct decisions versus incorrect decisions.

TASK	GOAL
1	Train and test the plain AI baseline
2	Build and check out an imperfect expert
3	Learn when to defer, using full expert data
4	Learn the expert’s strengths on a limited query budget

Table 1: Task list

Task 1 Baseline Classification

Experiment and Design Choice

We trained a TfidfVectorizer on the entire training set to feed into Logistic Regression. There was nothing complicated about it. We chose default word-unigram settings for the vectorizer, default L2 penalty, C = 1.0. We bumped max_iter up to 1000 just so it would actually converge. We skipped a validation split, hyperparameter search, bigrams, or any extra feature tuning. That was not the point of this task.

We picked this setup mainly because it is simple, fast on sparse text, and gives us class probabilities we can reuse in the deferral tasks later. Task 1 is only supposed to give us a reference point, not to achieve the highest possible accuracy.

Results

The model achieved 91.80% test accuracy. “Sports” was the class the model predicted correctly most often. “Business” was the class with the lowest recall rate. Most of the confusion occurred between the ‘Business’ and “Science/Technology” classes, which makes sense because the vocabularies of these two classes overlap significantly. The TF-IDF vectorizer resulted in 65,006 features.

CLASS	PRECISION	RECALL	F1	ARTICLES
WORLD	0.93	0.91	0.92	1,900
SPORTS	0.96	0.98	0.97	1,900
BUSINESS	0.89	0.88	0.89	1,900
SCI/TECH	0.89	0.90	0.90	1,900

Table 2: Task 1 classification results

TRUE CLASS	WORLD	SPORTS	BUSINESS	SCI/TECH
WORLD	1,724	52	73	51
SPORTS	18	1,863	13	6
BUSINESS	63	14	1,676	147
SCI/TECH	47	18	121	1,714

Table 3: Task 1 confusion matrix

Task 2 Simulated Expert

Experiment and Design Choice

For the expert, we made a sports specialist who is not particularly good at any other subject. An article counts as “inside the specialty” if it contains at least two different words from a fixed sports vocabulary list. Within this area, the expert answers 95% of the questions correctly; outside of it, only 55%. When the expert gives an incorrect answer, it makes a random selection from the other three categories.

We chose the keyword rule because it provides an area of expertise that is easy for the expert to define and measure. Requiring not just one, but two sports terms reduce the likelihood of random matches. Since the expert stays genuinely weak outside its specialty, the later tasks have something meaningful to learn when deferring is worth it and when it is not.

Results

MEASURE	RESULT
ACCURACY IN THE AREA OF SPORTS	97.41%
ACCURACY OUTSIDE THE AREA OF SPORTS	54.98%
OVERALL EXPERT ACCURACY	56.92%
SPECIALTY COVERAGE	4.57%
ARTICLES IN THE AREA	347 of 7,600

Table 4: Task 2 Simulated expert performance

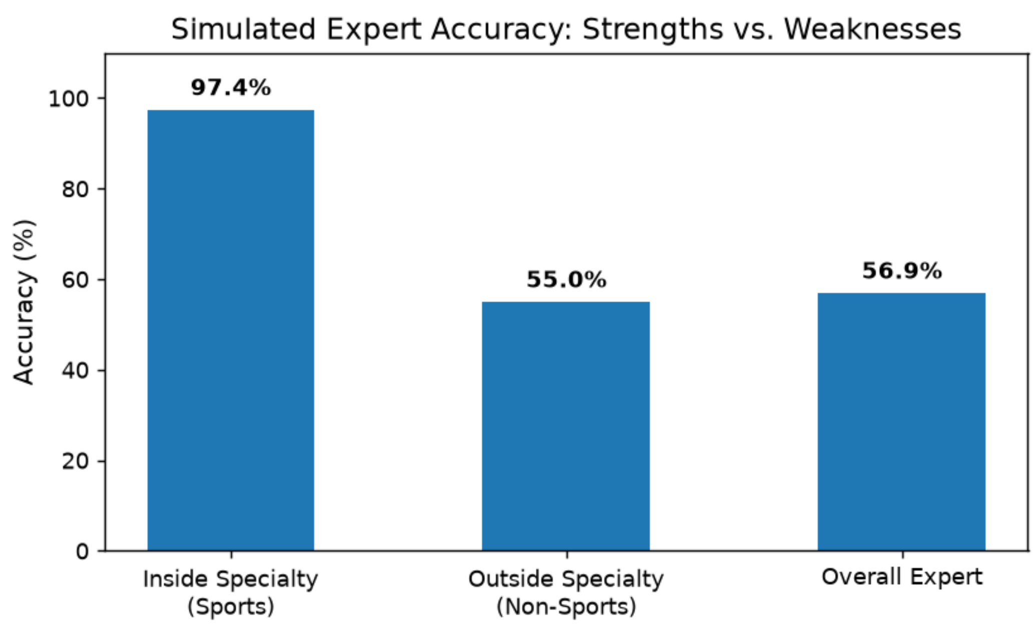

Figure 1: Expert accuracy inside and outside the sports specialty

The expert has truly outperformed AI in their field of expertise. The catch is that this field of expertise covers only 347 of the 7,600 test articles (4.57%); therefore, it makes sense to use the expert if the system can reliably determine which articles fall within this limited scope.

Task 3 Learning to Defer

Experiment and Design Choice

For Task 3, we assume we actually have the expert's answer for every training article. We marked each of those answers as right or wrong, then trained a second Logistic Regression model, reusing the same TF-IDF features from Task 1, to predict how likely the expert is to be correct on a given article. The system defers to the expert whenever that predicted probability beats the AI's own top class probability.

This compares the confidence level of the AI with the expert's predicted reliability and ensures that the higher one wins. Reusing TF-IDF meant there was no need for a second feature processing step. Test labels and the hidden specialty rule are only used to check how well we did afterward. They are never used to make the actual routing decision.

Results

MEASURE	RESULT
AI BASELINE ACCURACY	91.80%
TEAM ACCURACY	91.43%
GAIN OVER AI	-0.37 percentage points
DEFERRAL RATE	7.74%
BENEFICIAL DEFERRALS	19.73% of deferred cases
HARMFUL DEFERRALS	24.49% of deferred cases
NEUTRAL DEFERRALS	55.78% of deferred cases

Table 5 Task 3 team performance and deferral quality

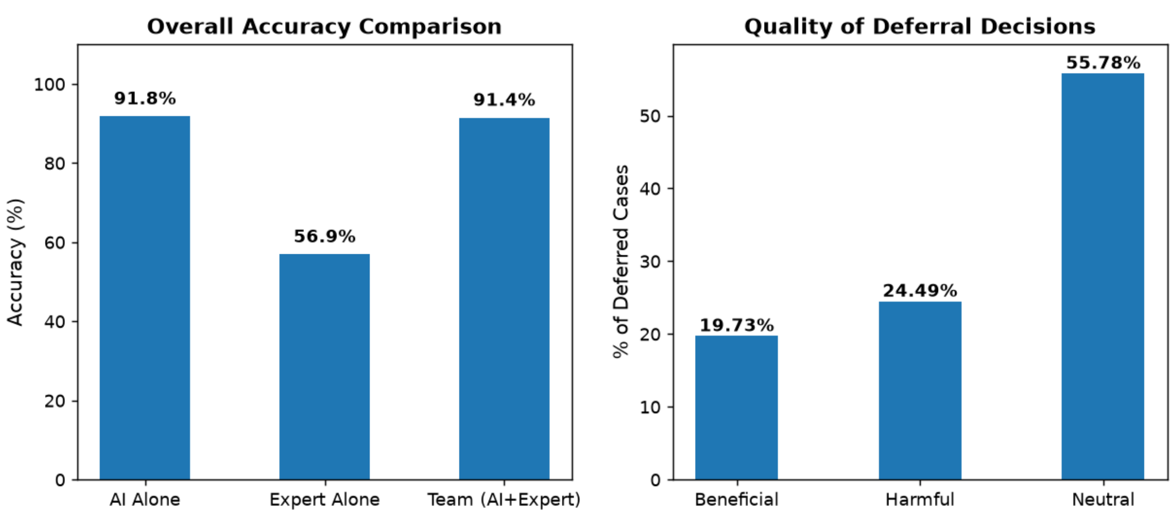

Figure 2 Task 3 accuracy and deferral quality

As a result, the team consisting of both the AI and the expert performed only 0.37 points lower than the result achieved by the AI alone. That is 28 correct predictions lost. Harmful deferrals outnumbered beneficial ones, which means the competence model was sometimes talking the AI out of a correct answer and handing the case to the expert instead, who then got it wrong. The remaining 55.78% were neutral deferrals where both systems had the same outcome: either both were correct or both were wrong.

Task 4 Active Learning

Experiment and Design Choice

This time we start with zero expert training data. We tried query budgets of 50, 100, 250, and 500. Random selection is our baseline for comparison. For active selection we used uncertainty sampling: asking the expert about the training articles where the AI is least confident. We then train the same kind of competence model as in Task 3, just on whatever answers we managed to collect, and apply the same deferral rule.

The idea behind uncertainty sampling is simple: spend your limited questions on the cases where the AI is uncertain. Comparing it to random selection tells us whether that focus works. To keep things fair, we reused the same simulated expert answers across every budget and strategy.

In the tables below, “Deferred” shows how often the system handed a test article to the expert. “Beneficial” and “Harmful” columns show, considering only the deferred cases, how often these forwarding decisions were helpful or harmful.

Random Selection Results

BUDGET	ACCURACY	GAIN PP	DEFERRED	BENEFICIAL	HARMFUL
50	90.97%	-0.83	10.88%	18.98%	26.60%
100	91.05%	-0.75	10.33%	19.24%	26.50%
250	90.96%	-0.84	11.18%	18.82%	26.35%
500	91.39%	-0.41	8.17%	20.45%	25.44%

Table 6 Random query results

Active Selection Results

BUDGET	ACCURACY	GAIN PP	DEFERRED	BENEFICIAL	HARMFUL
50	91.14%	-0.66	9.55%	19.01%	25.90%
100	90.92%	-0.88	10.99%	18.68%	26.71%
250	91.07%	-0.73	10.17%	18.76%	26.00%
500	91.17%	-0.63	8.84%	19.05%	26.19%

Table 7 Uncertainty query results

Competence Discovery Results

We also checked whether each strategy found the expert’s specialty and whether the competence model could rank correct expert answers above incorrect ones. Competence AUROC of 0.5 means roughly random ranking. These evaluation values were not used for selection or routing.

BUDGET	RANDOM SPECIALTY	UNCERTAINTY SPECIALTY	RANDOM AUROC	UNCERTAINTY AUROC	TEAM ACCURACY R / U
50	2 (4.00%)	0 (0.00%)	0.506	0.493	90.97% / 91.14%
100	4 (4.00%)	0 (0.00%)	0.506	0.493	91.05% / 90.92%
250	13 (5.20%)	1 (0.40%)	0.512	0.492	90.96% / 91.07%
500	27 (5.40%)	1 (0.20%)	0.512	0.488	91.39% / 91.17%

Table 8 Task 4 competence discovery results

We also tested two additional query strategies: class-balanced uncertainty sampling and sequential expert-competence uncertainty sampling. Neither provided a consistent improvement over the original uncertainty strategy in this fixed simulation. Class-balanced uncertainty produced results similar to the original uncertainty strategy, while expert-competence uncertainty achieved slightly higher AUROC but lower team accuracy. Therefore, the original uncertainty strategy is used for the main comparison.

Task 4 Results and Limitations

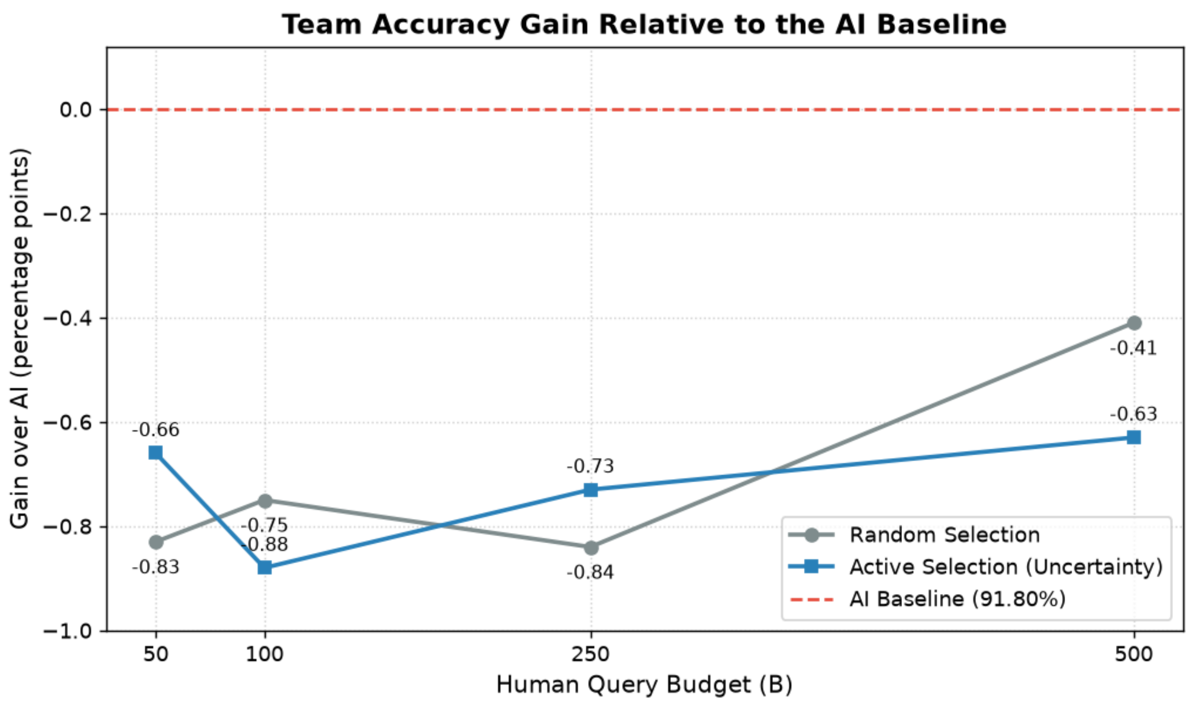

Figure 3 Team accuracy gain relative to the AI baseline by query budget

In this run, active selection produced higher accuracy at budgets 50 and 250, while random selection produced higher accuracy at 100 and 500. The best Task 4 result was 91.39%, from 500 random queries. Figure 3 shows every result relative to the 91.80% AI baseline, so zero means matching the AI and every negative value means losing accuracy. Nothing beat the baseline, and increasing the query budget did not consistently improve accuracy.

Part of the problem is that AI uncertainty does not necessarily point to the articles that would actually reveal the sports expert's specialty. Random selection found between 2 and 27 specialty articles as the budget increased. These made up 4.00% to 5.40% of the queried articles, close to the specialty's 4.41% coverage in the training set. Uncertainty selection found none at budgets 50 and 100, then only one at 250 and 500. Its competence AUROC stayed between 0.488 and 0.493, roughly random ranking. That explains why focusing on AI uncertainty did not reliably learn when this expert was useful.

Limitations

- The expert's specialty is defined by a keyword list we wrote by hand, so it's subjective.
- That specialty is rare in the data, which makes it a hard target to learn from.
- We compared AI confidence and expert competence directly, without calibrating either one, so the two probability estimates may not be directly comparable.
- We ran Tasks 3 and 4 with one fixed simulation, so the comparison describes this run rather than claiming that one strategy will generally perform better.

Conclusion

We got all four experiments up and running. TF-IDF and Logistic Regression provided us a 91.80% AI baseline. The sports expert we simulated achieved an accuracy rate of 97.41% within its area of expertise, but outside that area, the accuracy rate dropped to 54.98%. Task 3 learned a deferral rule from full expert data and reached 91.43% team accuracy. Task 4 learned from a limited number of expert queries and achieved an accuracy rate of 91.39%.

Neither deferral system managed to outperform the simple AI. The main reason for this was that harmful deferrals occurred more frequently than beneficial ones. Uncertainty sampling also failed to consistently outperform simple querying. These results suggest that our competence model and query strategy were not effective enough to improve on the strong AI baseline in this setting.

References

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